

Rural banks face a technology adoption gap

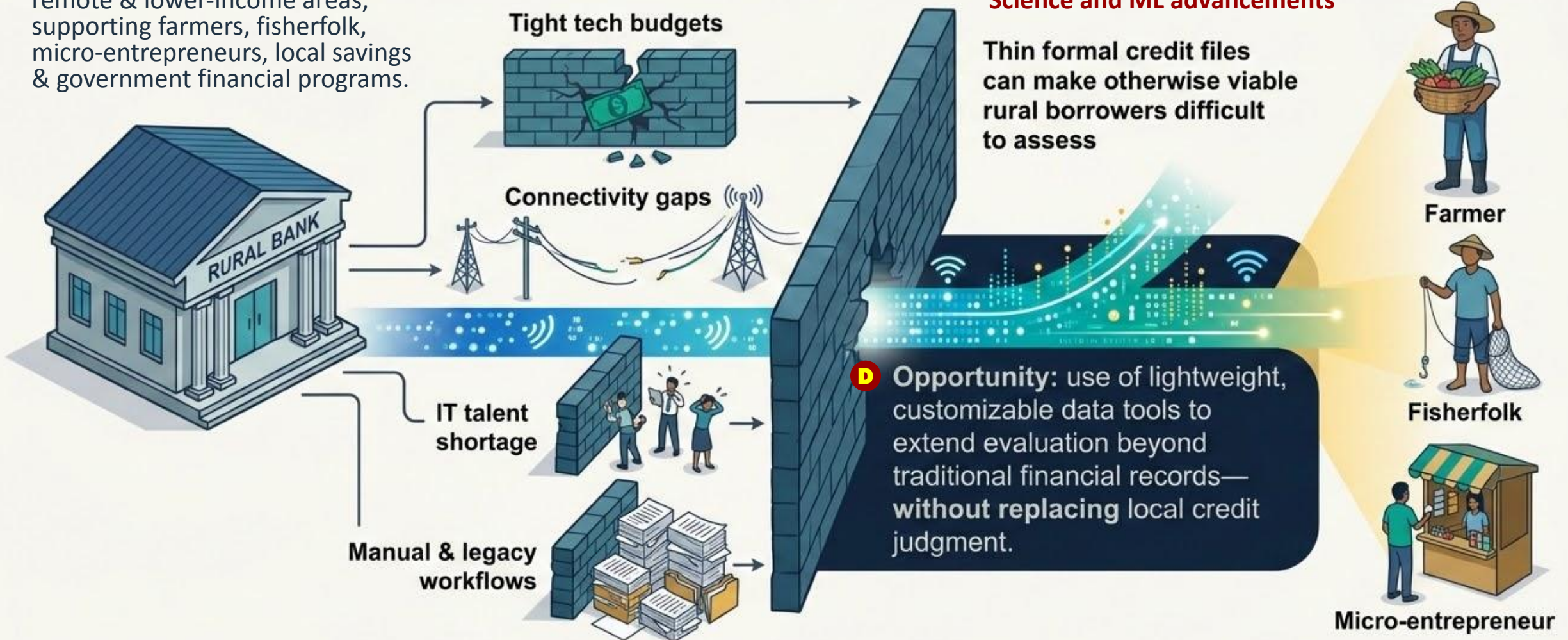
A Rural banks are **crucial last-mile financial institutions**, serving remote & lower-income areas, supporting farmers, fisherfolk, micro-entrepreneurs, local savings & government financial programs.

B Rural banks face daunting **Tech challenges** ...

C ... that prevent them from taking advantage of the latest **Data Science and ML advancements**

Thin formal credit files can make otherwise viable rural borrowers difficult to assess

D Opportunity: use of lightweight, customizable data tools to extend evaluation beyond traditional financial records—**without replacing local credit judgment.**



Machine learning can help close the information gap

Non-traditional indicators



Digital money

Mobile money use, digital transfers, merchant payments



Payment habits

Utilities and recurring large-purchase payments



Mobile engagement

Phone access, mobile use and top-up patterns



Capacity & resilience

Income quintile, emergency-fund access, work search

Alternative credit scoring



- a) New indicators can surface financial behaviors not visible in bureau or bank histories.
- b) designed as a lightweight proof of concept that can be rapidly customized to a bank's data, processes and risk appetite.

APPROVED

DECLINED

Business value

Responsibly approve viable applicants missed by conventional rules.

identify risks hidden by sparse traditional data.

Approval uplift can translate into portfolio growth



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₱737M

+ 10%
uplift

= ₱74Mn

Indicative average rural bank portfolio:
₱276.5B industry net loan portfolio ÷
~375 RCBs ≈ ₱737M per institution;
source: BSP/RBAP 2024 rural & coop. bank data

**Conservative uplift from alternative
credit scoring mechanism:**

- Machine learning may identify creditworthy applicants who fail conventional thresholds
- 15% to 47% historical increase with behavioral scoring *source: credolabs*

Indicative rural bank portfolio growth

BSP/RBAP industry totals are not a published median or typical individual-bank balance; per-bank figure is a rough sector-average proxy.

Better risk signals may reduce avoidable credit losses



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- 2 ppts

Conservative reduction in past due & NPLs:

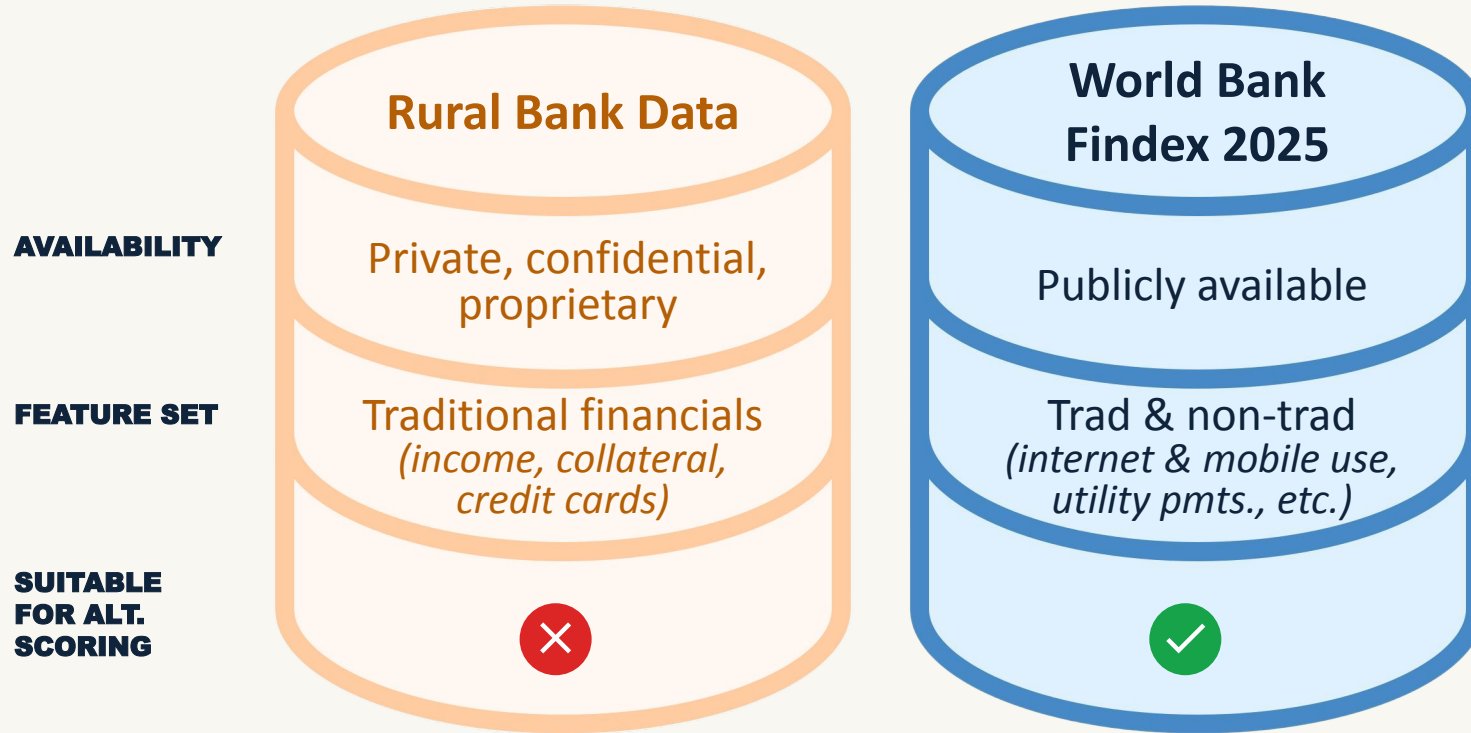
- Alternative features may reveal repayment capacity, financial behavior or stress signals conventional rules miss
- 10% to 15% historical RCB overdue or default rate; assume 2ppts reduction

= ₱15Mn

Indicative reduction in Past Due / NPLs

BSP reported a rural-bank-group past-due ratio of 11.31% in Jun-2026; past-due ratio is not the same as default rate.

Selecting a dataset for a prototype model



What is the Findex dataset?

- Global survey on financial inclusivity conducted by World Bank
- 144K total cases
- 9.7K records with loan application and approval data

Key limitations:

- Survey responses & approval flag—not verified repayment outcomes or default labels.
- Cross-country, self-reported data are not specific to Philippine rural-bank applicants.
- Feature availability, definitions and selection effects may not match bank operations.
- Findex has no direct rental-payment history; large-purchase regularity is only a proxy.

XGBoost algorithm with class-aware tuning performs best

EDA + Feature Engineering	Model Selection	Model Tuning	Best Model
● 9,691 applicant records; ● 78 retained features ● Imputation ● Rare-value grouping ● Chi-square screening ● One-hot encoding	Models evaluated and metric (declined recall): ● L1 (Lasso) Logistic Regression: 0.71 ● CatBoost: 0.72 ● XGBoost: 0.73 NOTE: Recall of minority class treated as target metric for prediction of applications to decline	● 3-fold CV with class-balanced weights ● Best parameters: ○ learning rate 0.01 ○ max_depth 5 ○ estimators 100	Tuned XGBoost selected: ● Precision 0.42 ● Recall 0.73 ● F1 Score 0.53 ● Accuracy 0.66 XGBoost’s lower precision implies a conservative bias with more false declines, prioritizing risk mitigation

Audit shows most impactful characteristics — and an inclusion trade-off



Review of our model's feature importance...

TOP CONTRIBUTORS TO APPROVE/DECLINE DECISION*

- Prior formal-institution borrowing
- Prior mobile-money borrowing
- Digital in-store payment frequency
- Emergency situation liquidity
- Top-quintile household income

** based on SHAP; LIME feature importance broadly aligned: prior borrowing dominated, with liquidity & mobile use, also prominent*



... reveal bias against unbanked individuals

MODEL LIMITATIONS

- Traditional credit-experience features can disadvantage unbanked and thin-file applicants and reinforce prior access.
- Global survey relationships may not transfer to local rural-bank borrowers.
- Proxy variable of loan approval as target & may not translate to repayment quality.
- Minority recall prioritized over precision; high rate of “false declines” expected.
- Black box nature of XGBoost.



A model variant was trained using **non-traditional features only** .

- “Decline” recall deteriorated from 0.73 to 0.61
- Feature importance shifted to alternative transactional, digital and behavioral trustworthiness indicators (bill payment via mobile, frequency of in-store merchant payments, frequency of deposits, phone & digital literacy).

Alternative XGBoost model provides **a financial inclusivity option**, but, **with a performance trade-off**

Move from global proof of concept to governed local pilot



1. Build a local outcome dataset

Link applications to repayment performance, delinquency, restructures and recoveries; define default consistently.



2. Co-design with loan officers

Map existing workflows, consent, data access, manual fallback and rural connectivity constraints.



3. Test features for coverage

Validate utility, mobile-money and digital-payment data availability; avoid penalizing people with no digital footprint.



4. Retrain & validate locally

Use time-based holdouts and out-of-time testing; compare baseline rules, logistic and tree models.



5. Govern decisions

Set human review, reason codes, override logs, privacy controls, security and model ownership.



6. Pilot safely & monitor

Shadow-score first; then controlled rollout with exposure caps, fairness metrics, drift and vintage tracking.



DECISION GATE: expand only when local repayment outcomes, fairness checks & operational controls meet bank-approved thresholds.